Overlap of stool-decreased cellulose-annotated taxa against six null models

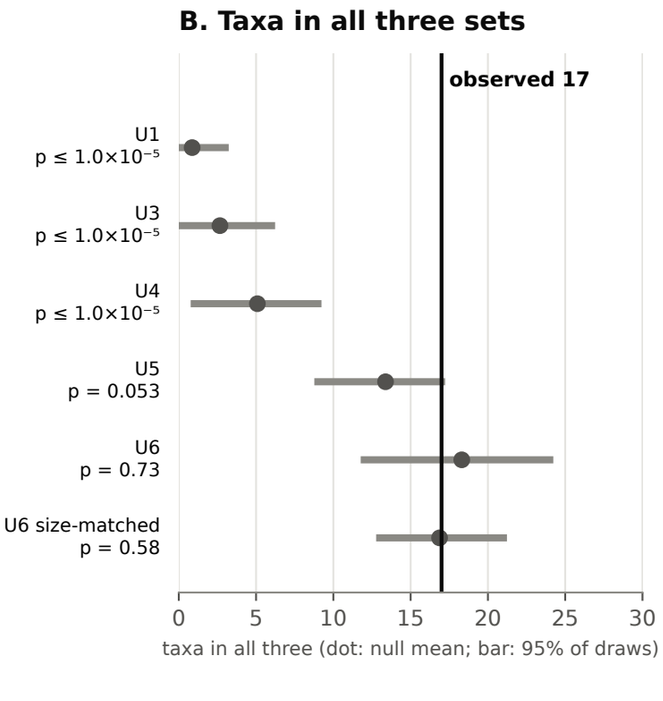
Nulls that draw taxa independently (U1–U4) make the overlap look extreme; nulls that respect how often taxa are reported (U5) and which taxa are reported together (U6) do not.

A. Expected Jaccard distance and p under each null

	observed d_J	U1 matrix uniform	U2 stool uniform	U3 stool dec uniform	U4 random trait	U5 prevalence weighted	U6 signature permutation	U6 size-matched
PD vs. HIV	**0.488**	0.914 exact 1.1×10^{-14}	0.876 exact 1.8×10^{-11}	0.841 exact 2.5×10^{-9}	0.795 p 3.0×10^{-5}	0.621 p 0.024	0.550 p 0.21	0.541 p 0.20
PD vs. CRC	**0.400**	0.894 exact 1.0×10^{-17}	0.845 exact 1.6×10^{-13}	0.799 exact 1.1×10^{-10}	0.747 p $\le 1.0\times10^{-5}$	0.546 p 0.019	0.491 p 0.10	0.462 p 0.18
HIV vs. CRC	**0.550**	0.918 exact 2.6×10^{-11}	0.882 exact 1.2×10^{-8}	0.848 exact 6.8×10^{-7}	0.783 p 0.001	0.622 p 0.19	0.540 p 0.55	0.537 p 0.58
Mean of 3 pairs	**0.479**	0.909 p $\le 1.0\times10^{-5}$	0.868 p $\le 1.0\times10^{-5}$	0.829 p $\le 1.0\times10^{-5}$	0.775 p $\le 1.0\times10^{-5}$	0.596 p 0.005	0.527 p 0.19	0.513 p 0.24

Observed values are in the first column. Shaded blue: $p < 0.05$. U1–U3 pair p-values are exact hypergeometric tails; all other p are Monte Carlo. U4 places a random trait on the taxa reported decreased in stool in the three conditions; U5 weights taxa by their reporting frequency in other stool studies; U6 permutes whole study signatures.

B. Taxa in all three sets



C. The 47 taxa by the conditions that report them decreased in stool

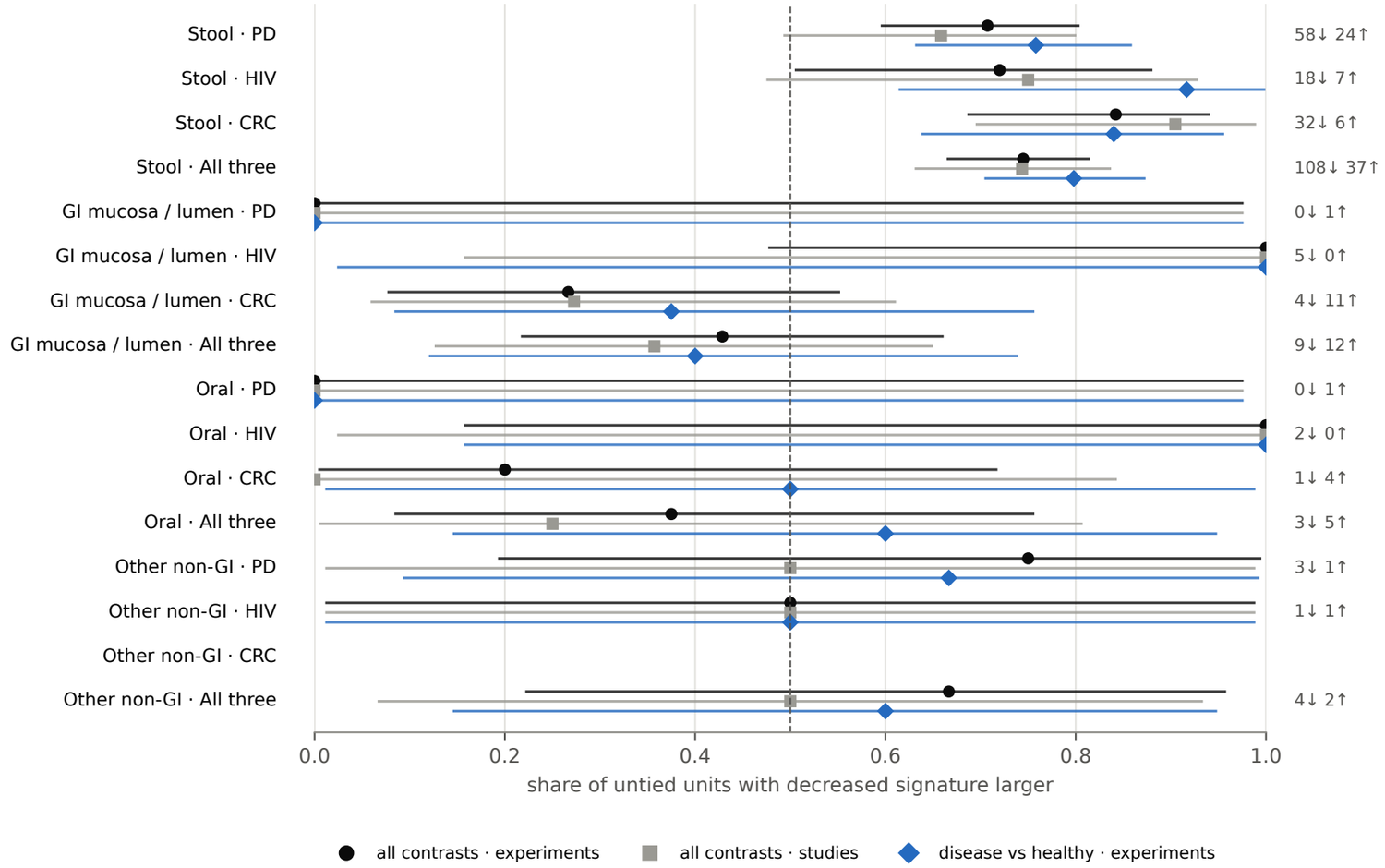
PD, HIV and CRC 17	PD and CRC only 10	PD and HIV only 4	HIV and CRC only 1	PD only 7	CRC only 6	HIV only 2
Lachnospiraceae	Roseburia faecis	Bacteroidales	Actinomycetota	Eubacteriaceae	Blautia obeum	Bacillus
Roseburia	Anaerostipes hadrus	Bacteroidaceae		Gammaproteobacteria	Hominimerdicola alba	Carnobacteriaceae
Blautia	Bacilli	Lachnoclostridium		Lacrimispora	Lachnospira multipara	
Clostridium	Comamonadaceae	Mediterraneibacter torques		Mediterraneibacter	Pseudomonadaceae	
Bacillota	Lactobacillales			Micrococcaceae	Ruminococcus champanellensis	
Bacteroides	Bacteroides cellulosilyticus			Sphingobacteriaceae	Serratia	
Clostridia	Pseudomonadota			Streptomycetaceae		
Lachnospira	Ruminococcus bromii					
Oscillospiraceae	Ruminococcus flavefaciens					
Ruminococcus	[Eubacterium] siraeum					
Clostridiaceae						
Eubacteriales						
Anaerostipes						
Bacteroidia						
Bacteroidota						
Eubacterium						
Butyrivibrio						

Names are KG-Microbe matrix taxon names; family-and-above nodes carry annotations rolled up from descendants. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.

Direction of the cellulose annotation shift by body site, inference unit and contrast type

Share of untied units whose decreased signature had the larger cellulose annotation share, with exact 95% intervals; 0.5 means no preference.

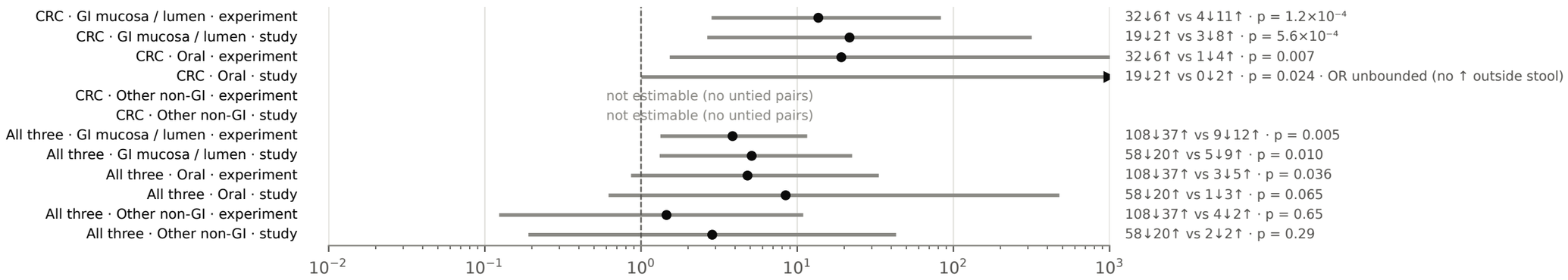
A. Share of untied units with the decreased signature larger



B. Set definitions: pairwise and mean d_J

	PD–HIV	PD–CRC	HIV–CRC	mean	core/union
Primary: all ranks	0.488	0.400	0.550	**0.479**	17/47
Genus and species ranks only	0.450	0.375	0.609	**0.478**	9/25
Profiles with cellulose z > 0 only	0.474	0.477	0.538	**0.496**	16/46
Disease-vs-healthy contrasts only	0.556	0.450	0.545	**0.517**	13/41
Disease-vs-healthy, strong evidence	0.556	0.450	0.545	**0.517**	13/41
All body sites, not only stool	0.551	0.446	0.588	**0.529**	19/60
Net decreased (more dec than inc profiles)	0.645	0.622	0.657	**0.641**	7/40
Net decreased (more dec than inc studies)	0.586	0.667	0.676	**0.643**	7/39
Baseline: all stool-decreased taxa, any trait	0.795	0.747	0.783	**0.775**	51/471

C. Stool against each other compartment (odds ratio of the ↓ share, exact 95% interval)



Experiments within a study are not independent; the study rows collapse them to one vote per study. Disease-vs-healthy rows keep experiments the contrast manifest classifies disease_vs_healthy, relabelled so decreased means lower in disease. Intervals are exact (Clopper–Pearson for shares; conditional maximum likelihood for odds ratios). PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.

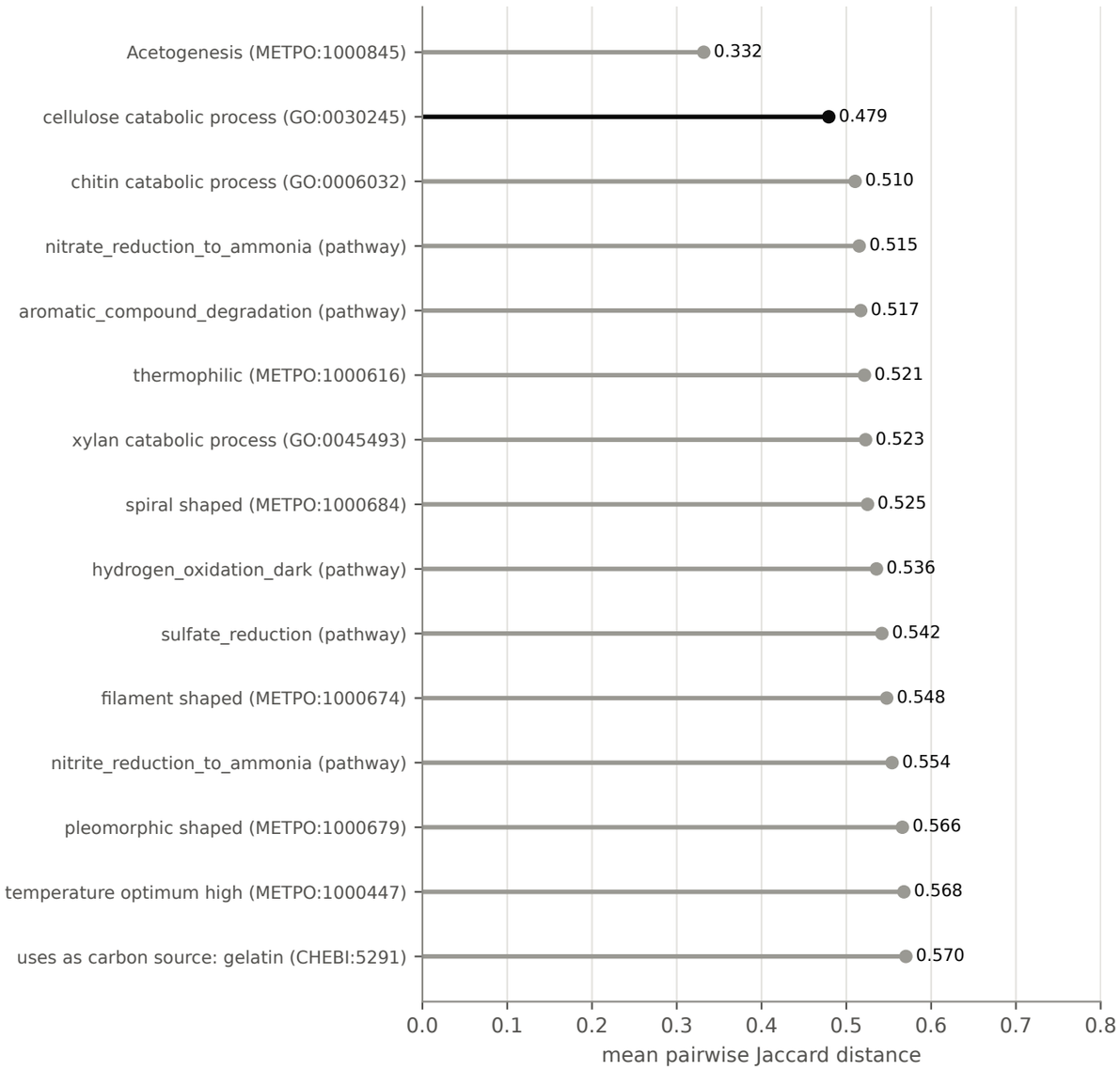
Taxa shared by at least two conditions, and the trait-wide convergence benchmark

32 cellulose-annotated taxa are reported decreased in stool in two or more conditions; cells give decreased/increased stool studies.

A. Decreased / increased stool studies

	PD	HIV	CRC	flags
Lachnospiraceae (family)	20/7	5/3	9/5	core · net↓
Roseburia (genus)	20/4	2/0	6/0	core · g/s · net↓
Blautia (genus)	13/2	1/3	6/1	core · g/s
Clostridium (genus)	7/3	5/2	6/3	core · g/s · net↓
Bacillota (phylum)	8/5	4/2	4/1	core · net↓
Bacteroides (genus)	6/3	8/0	2/2	core · g/s
Clostridia (class)	5/7	4/0	7/1	core
Lachnospira (genus)	5/2	3/0	6/1	core · g/s · net↓
Oscillospiraceae (family)	8/13	4/0	2/1	core
Ruminococcus (genus)	6/2	4/0	4/3	core · g/s · net↓
Clostridiaceae (family)	8/2	3/0	2/2	core
Eubacteriales (order)	5/6	4/1	4/1	core
Anaerostipes (genus)	2/1	3/0	3/1	core · g/s · net↓
Bacteroidia (class)	5/2	2/0	1/4	core
Bacteroidota (phylum)	3/0	3/1	1/4	core
Eubacterium (genus)	1/1	1/1	3/2	core · g/s
Butyrivibrio (genus)	2/2	1/2	1/0	core · g/s
Bacteroidales (order)	4/1	3/0	0/4	
Roseburia faecis (species)	5/0	0/0	2/0	g/s
Bacteroidaceae (family)	1/2	5/0	0/2	
Actinomycetota (phylum)	0/7	2/1	2/1	
Anaerostipes hadrus (species)	1/0	0/0	2/1	g/s
Bacilli (class)	2/2	0/2	1/2	
Comamonadaceae (family)	2/0	0/0	1/0	
Lactobacillales (order)	2/1	0/1	1/2	
Bacteroides cellulosilyticus (species)	1/1	0/0	1/0	g/s
Lachnoclostridium (genus)	1/1	1/1	0/1	g/s
Mediterraneibacter torques (species)	1/3	1/3	0/3	g/s
Pseudomonadota (phylum)	1/9	0/2	1/3	
Ruminococcus bromii (species)	1/2	0/0	1/0	g/s
Ruminococcus flavefaciens (species)	1/0	0/0	1/0	g/s
[Eubacterium] siraeum (species)	1/1	0/0	1/0	g/s

B. Fifteen lowest of 142 non-MediaDive traits



Panel A fill: blue where decreased studies outnumber increased, red where the reverse, darker with a larger difference. core = in all three sets; g/s = genus or species rank; net↓ = more decreased than increased studies in all three conditions.

Trait names come from taxa_trait_matrix_columns.tsv (object_name). Family-and-above nodes carry rolled-up annotations. PD, Parkinson's disease; HIV, HIV infection; CRC, colorectal cancer. BugSigDB 2026-04-24_00:41_UTC; human profiles; condition and body site from each experiment's own record.